



# Elastic analysis for thick cylinders and spherical pressure vessels made of functionally graded materials

Y.Z. Chen<sup>\*</sup>, X.Y. Lin

Division of Engineering Mechanics, Jiangsu University, Zhenjiang, Jiangsu 212013, PR China

## ARTICLE INFO

### Article history:

Received 8 April 2007

Received in revised form 19 April 2008

Accepted 22 April 2008

Available online 9 June 2008

### Keywords:

Functionally graded materials

Elastic analysis

Numerical technique

Thick cylinder

Spherical pressure vessel

## ABSTRACT

In this paper, elastic analysis for a thick cylinder made of functionally graded materials (FGMs) is carried out. The property of FGMs is assumed to be exponential function form. The problem is reduced to solve an ordinary differential equation numerically. Stress distributions along the radial direction are studied. The obtained result shows that the property of FGMs has a significant influence to the stress distribution along the radial direction. Numerical results are given which are useful for engineer to design a cylinder made of FGMs. Similar analysis for spherical pressure vessel made of FGMs is presented.

© 2008 Elsevier B.V. All rights reserved.

## 1. Introduction

From viewpoints of solid mechanics, the functionally graded materials (FGMs) are non-homogeneous elastic mediums. Recently, investigation for FGMs has received considerable attention by many researchers. The nonlinear bending analysis of a plate with FGMs property was investigated [1]. Crack problems for elastic medium made of FGMs were investigated by several investigators [2–4].

Mechanical and thermal stresses in a functionally graded hollow cylinder due to radially symmetric loads were studied [5]. In the paper, the modulus of elasticity is assumed in the form of power law, and a closed form solution was obtained. If the modulus is not in the form of power law, the problem will be more complicated. Elastic analysis of internally pressurized thick-walled spherical pressure vessels of functionally graded materials was studied [6]. In the paper, the modulus of elasticity is assumed in the form of exponential function. Since the solution for a stress component was obtained in the series form, one should truncate several terms to obtain an approximate solution.

For a given loading, if comparing with the homogenous materials the stress distribution is significantly changed in the FGMs case. Using this property in the FGMs, engineer can design composite materials such that any portion of materials reaches the same safety level.

The thick cylinder is a pressure vessel commonly used in industry. From the strength analysis it is seen that the most dangerous point is located at the inner portion of the cylinder, if the homogeneous materials are used. However, if one uses the FGMs for a thick cylinder, this situation can be changed. This point can be found from the following analysis.

In this paper, the elastic analysis for a cylinder made of FGMs is carried out. The problem is reduced to solve an ordinary differential equation numerically. The obtained result shows that the FGMs property has a significant influence to the stress distribution. Numerical results are given which are useful for engineer to design a cylinder made of FGMs. Similar analysis for spherical pressure vessel made of FGMs is present.

## 2. Elastic analysis for a thick cylinder made of functionally graded material

A long cylinder with inner radius “ $a$ ” and outer radius “ $b$ ” is investigated. The cylinder is assumed under the action of an inner pressure  $q_a$  or a loading  $q_b$  on the outer circular boundary (Fig. 1). The problem can be studied in polar coordinates  $(r, \theta)$ . In the symmetrical deformation case, the one displacement “ $u$ ” is in the  $r$ -direction. Two strain components can be expressed as [7–9]

$$\varepsilon_r = \frac{du}{dr}, \quad \varepsilon_\theta = \frac{u}{r} \quad (1)$$

From Eq. (1), the compatibility condition of displacement will be

<sup>\*</sup> Corresponding author. Tel.: +86 511 88780780; fax: +86 511 88791739.  
E-mail address: [chens@ujs.edu.cn](mailto:chens@ujs.edu.cn) (Y.Z. Chen).

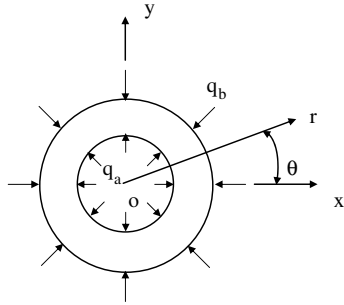


Fig. 1. A cylinder with inner pressure  $q_a$  or a loading  $q_b$  at outer boundary.

$$\varepsilon_r = \frac{d(r\varepsilon_\theta)}{dr} \quad \text{or} \quad \varepsilon_r = \varepsilon_\theta + r \frac{d\varepsilon_\theta}{dr} \quad (2)$$

For the stress components  $\sigma_r$  and  $\sigma_\theta$ , the equilibrium equation takes the form

$$\frac{d\sigma_r}{dr} + \frac{\sigma_r - \sigma_\theta}{r} = 0 \quad (3)$$

Eq. (3) can be satisfied automatically, if one introduces a function  $F(r)$  and let

$$\sigma_r = \frac{F(r)}{r}, \quad \sigma_\theta = \frac{dF}{dr} \quad (4)$$

In this paper, it is assumed that the Poisson's ratio  $\nu$  ( $=0.3$ ) takes a constant value, and the Young's modulus has the property [4]

$$E(r) = E_0 \exp\left(\frac{\beta(r-a)}{b-a}\right) \quad \text{with} \quad E(r)|_{r=a} = E_0, \quad E(r)|_{r=b} = E_0 \exp(\beta) \quad (5)$$

where  $E_0$  is a constant, and  $\beta$  takes negative or positive and represents the property of the FGMs, which is called the material parameter hereafter.

The material properties of FGM are continuous functions of position. Generally, the material coefficients of FGM manufactured in perfect process are changed in space without acuity. From Eq. (5), we can define a ratio by  $\gamma = E(r)|_{r=b}/E(r)|_{r=a}$  or  $\gamma = \exp(\beta)$ . The  $\gamma$  value represents the ratio of the Young's modulus at the outer boundary and the inner boundary. If  $\beta = 0, 0.1, 0.2$  and  $1$ , we have  $\gamma = 1, 1.1052, 1.2214$  and  $2.7183$ , respectively. Therefore, Eq. (5) can model the material properties of FGM when the properties are changed not significantly along the radial direction.

In the plane strain case, the stress-strain relation will be

$$\varepsilon_r = \frac{1-\nu^2}{E(r)} \left( \sigma_r - \frac{\nu}{1-\nu} \sigma_\theta \right), \quad \varepsilon_\theta = \frac{1-\nu^2}{E(r)} \left( \sigma_\theta - \frac{\nu}{1-\nu} \sigma_r \right) \quad (6)$$

Substituting Eq. (4) into (6), and then (6) into (2) yields

$$\frac{d^2 F}{dr^2} + \frac{1}{r} \frac{dF}{dr} - \frac{F}{r^2} - \left( \frac{dF}{dr} - \frac{\nu}{1-\nu} \frac{F}{r} \right) \frac{1}{E(r)} \frac{dE(r)}{dr} = 0 \quad (\text{or } A(F(r)) = 0) \quad (7)$$

Since the function  $E(r)$  is shown by Eq. (5), we have:

$$\frac{1}{E(r)} \frac{dE(r)}{dr} = \frac{\beta}{b-a} \quad (8)$$

Two boundary value problems are studied below.

In the first problem, the cylinder is subject to an inner pressure  $q_a$  and the boundary conditions take the following form (Fig. 1)

$$\sigma_r|_{r=a} = -q_a, \quad \sigma_r|_{r=b} = 0 \quad (\text{or } F|_{r=a} = -q_a a, \quad F|_{r=b} = 0) \quad (9)$$

The solution under the condition (9) can be sought in the form

$$F(r) = F_1(r) + cF_2(r) \quad (10)$$

In Eq. (10), two functions  $F_1(r)$  and  $F_2(r)$  are governed by

$$A(F_1(r)) = 0, \quad F_1|_{r=a} = -q_a a, \quad \frac{dF_1}{dr}|_{r=a} = 0 \quad (11)$$

$$A(F_2(r)) = 0, \quad F_2|_{r=a} = 0, \quad \frac{dF_2}{dr}|_{r=a} = 1 \quad (12)$$

where the operator  $A(F(r))$  has been indicated in Eq. (7). The two functions,  $F_1(r)$  and  $F_2(r)$  can be evaluated by using a numerical integration for which the Runge-Kutta method is used [10]. Thus, from the second condition in Eq. (9), we have

$$F_1(b) + cF_2(b) = 0 \quad \text{or} \quad c = -F_1(b)/F_2(b) \quad (13)$$

Therefore, the solution of the function  $F(r)$  can be obtained from Eq. (10).

In the solution of the differential equation,  $N = 200$  divisions are used [10]. The calculated results for stresses can be expressed as

$$\sigma_r = -f_1(a/b, \beta, r) q_a, \quad \sigma_\theta = f_2(a/b, \beta, r) q_a \quad (14)$$

The calculated results for two cases,  $a/b = 0.5$  and  $0.9$  are plotted in Figs. 2–5, respectively.

Five values for  $\beta$  are assumed,  $\beta = -2, -1, 0, 1$  and  $2$ , which corresponds to  $E(b)/E(a) = 0.1353, 0.3679, 1, 2.7183$  and  $7.3891$ , respectively. For the case of  $a/b = 0.5$ , the calculated results for  $f_1(a/b, \beta, r)$  and  $f_2(a/b, \beta, r)$  are listed in Tables 1 and 2, respectively.

In the homogeneous material case ( $\beta = 0, E(r) = E_0$ ), the available solution is [7–9]

$$\sigma_r = -\frac{(b^2 - r^2)a^2}{(b^2 - a^2)r^2} q_a, \quad \sigma_\theta = \frac{(b^2 + r^2)a^2}{(b^2 - a^2)r^2} q_a \quad (15)$$

From the plotted results or Tables 1 and 2, we see that the material parameter  $\beta$  can significantly affects the stress distribution along the radius direction. For example, in the case of  $a/b = 0.5$  and the homogeneous material ( $\beta = 0$  or  $E(r) = E_0$ ), the non-dimensional stresses  $f_2$  (the  $\sigma_\theta$  component) are changed from 1.667, 1.435, ... to 0.667, for  $(r-a)/(b-a) = 0, 0.1, \dots$  to 1.0, respectively (Fig. 3, the curve for  $\beta = 0$ ). However, in the case of  $\beta = 1$ , or  $E(b)/E(a) = 2.7183$ , the non-dimensional stresses  $f_2$  (the  $\sigma_\theta$  component) are varying within the range 0.933–1.171, nearly to a uniform distribution along the section (Fig. 3, the curve for  $\beta = 1$ ).

In the second problem, the cylinder is subject to a loading  $q_b$  at outer boundary (Fig. 1), and the boundary conditions take the following form

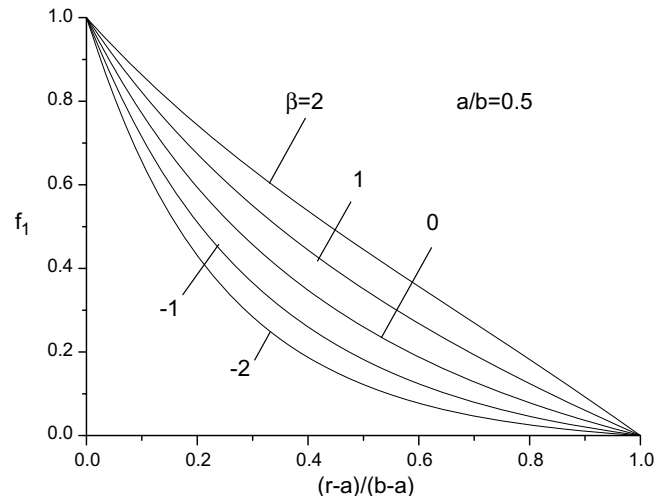
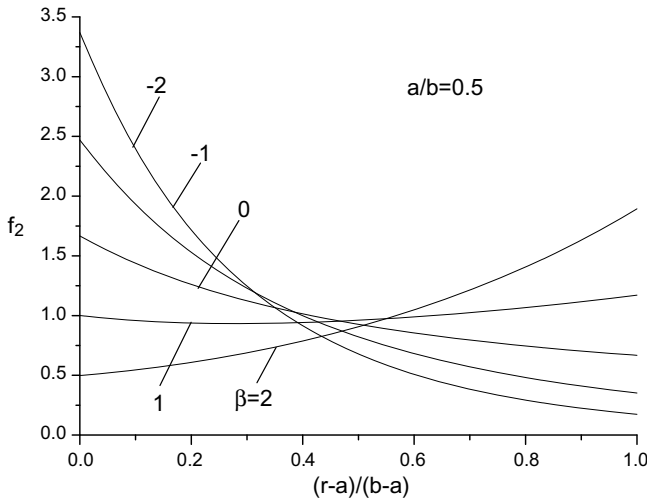
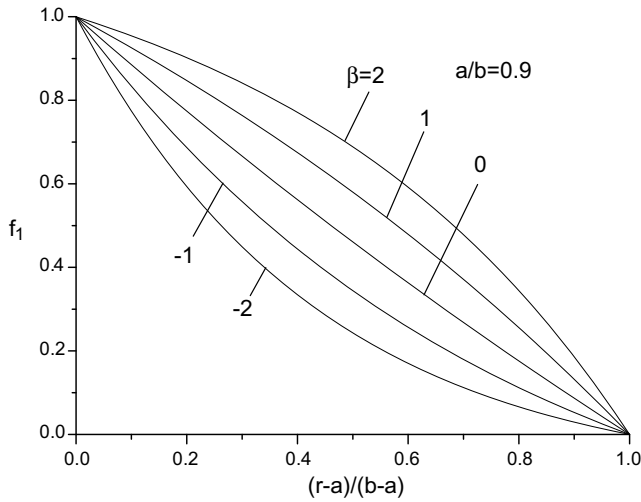


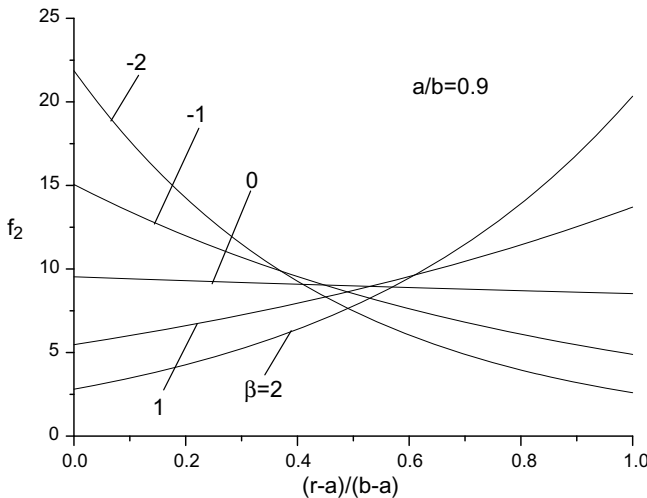
Fig. 2. Non-dimensional radial stress  $f_1(a/b, \beta, r)$ , the  $\sigma_r$  component in cylinder ( $a/b = 0.5$ , with an inner pressure  $q_a$ ).



**Fig. 3.** Non-dimensional tangential stress  $f_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in cylinder ( $a/b = 0.5$ , with an inner pressure  $q_a$ ).



**Fig. 4.** Non-dimensional radial stress  $f_1(a/b, \beta, r)$ , the  $\sigma_r$  component in cylinder ( $a/b = 0.9$ , with an inner pressure  $q_a$ ).



**Fig. 5.** Non-dimensional tangential stress  $f_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in cylinder ( $a/b = 0.9$ , with inner pressure  $q_a$ ).

$$\sigma_r|_{r=a} = 0, \quad \sigma_r|_{r=b} = -q_b \quad (\text{or } F|_{r=a} = 0, \quad F|_{r=b} = -q_b b) \quad (16)$$

The solution for the condition (15) can be sought in the form

$$F(r) = cF_1(r) \quad (17)$$

In Eq. (17), the function  $F_1(r)$  is defined by

$$\mathcal{A}(F_1(r)) = 0, \quad F_1|_{r=a} = 0, \quad \left. \frac{dF_1}{dr} \right|_{r=a} = 1 \quad (18)$$

where the operator  $\mathcal{A}(F(r))$  has been indicated in Eq. (7). The function  $F_1(r)$  can be evaluated by using a numerical integration for which the Runge–Kutta method is used [10]. Thus, from the second condition in Eq. (16), we have

$$cF_1(b) = -q_b b \quad \text{or} \quad c = -q_b b / F_1(b) \quad (19)$$

Therefore, the solution of the function  $F(r)$  can be obtained from Eq. (17).

Similarly, the calculated results for stresses can be expressed as

$$\sigma_r = -g_1(a/b, \beta, r)q_b, \quad \sigma_\theta = -g_2(a/b, \beta, r)q_b \quad (20)$$

The calculated results for two cases,  $a/b = 0.5$  and  $0.9$  are plotted in Figs. 6–9, respectively.

In the homogeneous material case ( $\beta = 0$ ,  $E(r) = E_0$ ), the available solution was obtained previously [7–9]

$$\sigma_r = -\frac{(r^2 - a^2)b^2}{(b^2 - a^2)r^2}q_b, \quad \sigma_\theta = -\frac{(r^2 + a^2)b^2}{(b^2 - a^2)r^2}q_b \quad (21)$$

From the plotted results, we see that the material parameter  $\beta$  can significantly affects the stress distribution along the radius direction. For example, in the case of  $a/b = 0.9$  and the homogeneous material ( $\beta = 0$  or  $E(r) = E_0$ ), the non-dimensional stresses  $g_2$  (the  $\sigma_\theta$  component) are varying within the range 10.526–9.526 (Fig. 9, the curve for  $\beta = 0$ ). However, in the case of  $\beta = 1$ , or  $E(b)/E(a) = 2.7183$ , the non-dimensional stresses  $g_2$  (the  $\sigma_\theta$  component) are changed from 6.233, 6.804, ... to 15.036, for  $(r - a)/(b - a) = 0, 0.1, \dots$  to 1.0, respectively. In this case, the outer portion is more rigid and the non-dimensional stress at outer portion is higher,  $g_2 = 15.036$  (Fig. 9, the curve for  $\beta = 1$ ).

### 3. Elastic analysis for a spherical pressure vessel made of functionally graded material

A spherical pressure vessel with inner radius “ $a$ ” and outer radius “ $b$ ” is investigated. The spherical pressure vessel is assumed under the action of an inner pressure  $q_a$  or a loading  $q_b$  on the outer circular boundary. The problem can be studied in spherical coordinates  $(r, \theta, \phi)$  [7–9]. In the symmetrical deformation case, the only displacement “ $u$ ” is in the  $r$ -direction. Two strain components can be expressed as [7–9]

$$\varepsilon_r = \frac{du}{dr}, \quad \varepsilon_\theta = \varepsilon_\phi = \frac{u}{r} \quad (22)$$

As before, from Eq. (22), the compatibility condition of displacement will be

$$\varepsilon_r = \frac{d(r\varepsilon_\theta)}{dr} \quad \text{or} \quad \varepsilon_r = \varepsilon_\theta + r \frac{d\varepsilon_\theta}{dr} \quad (23)$$

For the stress components  $\sigma_r$  and  $\sigma_\theta$ , the equilibrium equation takes the form

$$\frac{d\sigma_r}{dr} + \frac{2(\sigma_r - \sigma_\theta)}{r} = 0 \quad (24)$$

Eq. (24) can be satisfied automatically, if one introduce a function  $F(r)$  and let

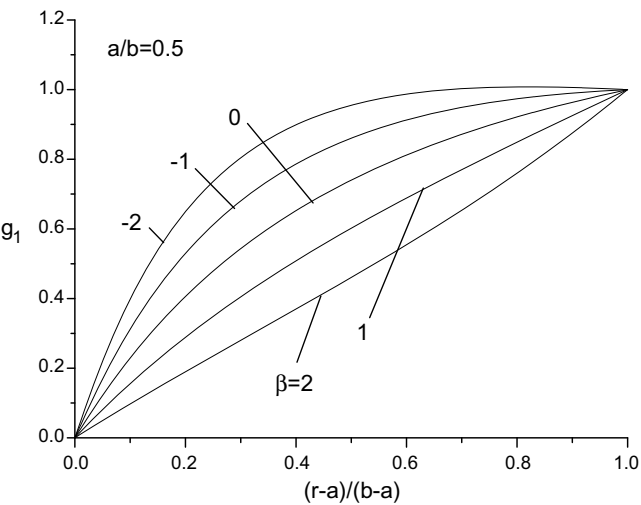
$$\sigma_r = \frac{F(r)}{r^2}, \quad \sigma_\theta = \frac{1}{2r} \frac{dF}{dr} \quad (25)$$

**Table 1**  
Non-dimensional radial stress  $f_1(a/b, \beta, r)$ , the  $\sigma_r$  component in cylinder ( $a/b = 0.5$ , inner pressure  $q_a$ )

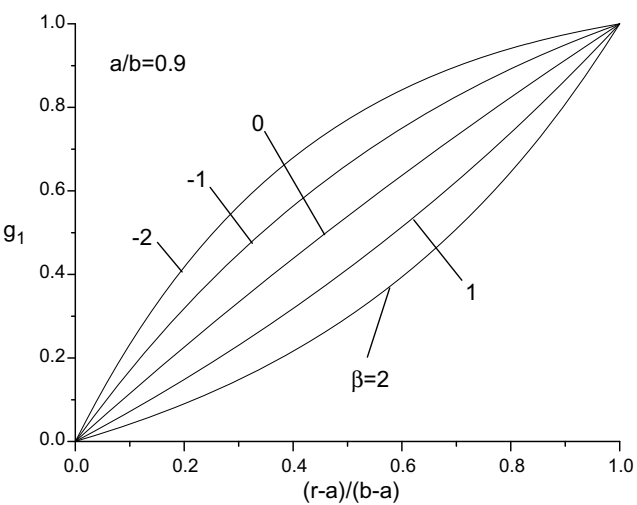
| $(r-a)/(b-a)$ | 0.000 | 0.100 | 0.200 | 0.300 | 0.400 | 0.500 | 0.600 | 0.700 | 0.800 | 0.900 | 1.000 |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $\beta = -2$  | 1.000 | 0.651 | 0.428 | 0.282 | 0.186 | 0.121 | 0.077 | 0.046 | 0.025 | 0.010 | 0.000 |
| $\beta = -1$  | 1.000 | 0.711 | 0.508 | 0.364 | 0.259 | 0.181 | 0.123 | 0.079 | 0.046 | 0.020 | 0.000 |
| $\beta = 0$   | 1.000 | 0.769 | 0.593 | 0.456 | 0.347 | 0.259 | 0.188 | 0.128 | 0.078 | 0.036 | 0.000 |
| $\beta = 1$   | 1.000 | 0.820 | 0.673 | 0.550 | 0.443 | 0.350 | 0.268 | 0.193 | 0.124 | 0.060 | 0.000 |
| $\beta = 2$   | 1.000 | 0.862 | 0.742 | 0.636 | 0.538 | 0.445 | 0.357 | 0.269 | 0.182 | 0.093 | 0.000 |

**Table 2**  
Non-dimensional tangential stress  $f_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in cylinder ( $a/b = 0.5$ , inner pressure  $q_a$ )

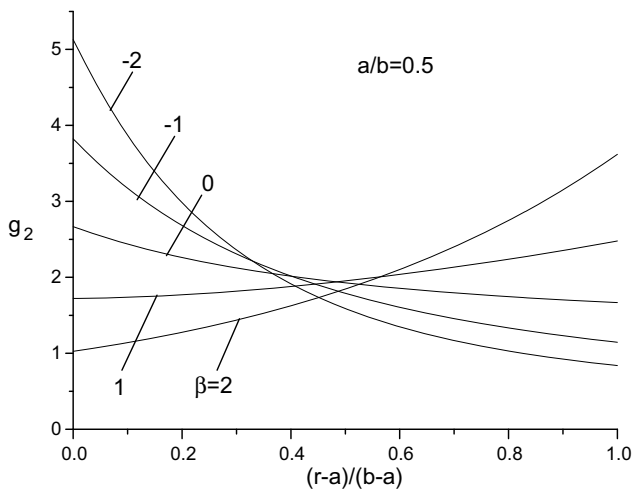
| $(r-a)/(b-a)$ | 0.000 | 0.100 | 0.200 | 0.300 | 0.400 | 0.500 | 0.600 | 0.700 | 0.800 | 0.900 | 1.000 |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| $\beta = -2$  | 3.370 | 2.382 | 1.709 | 1.241 | 0.912 | 0.676 | 0.506 | 0.382 | 0.291 | 0.224 | 0.174 |
| $\beta = -1$  | 2.470 | 1.926 | 1.526 | 1.226 | 0.997 | 0.820 | 0.681 | 0.570 | 0.482 | 0.410 | 0.352 |
| $\beta = 0$   | 1.667 | 1.435 | 1.259 | 1.122 | 1.014 | 0.926 | 0.854 | 0.795 | 0.745 | 0.703 | 0.667 |
| $\beta = 1$   | 1.001 | 0.958 | 0.937 | 0.933 | 0.942 | 0.961 | 0.988 | 1.024 | 1.066 | 1.115 | 1.171 |
| $\beta = 2$   | 0.498 | 0.543 | 0.607 | 0.688 | 0.787 | 0.906 | 1.047 | 1.212 | 1.405 | 1.630 | 1.893 |



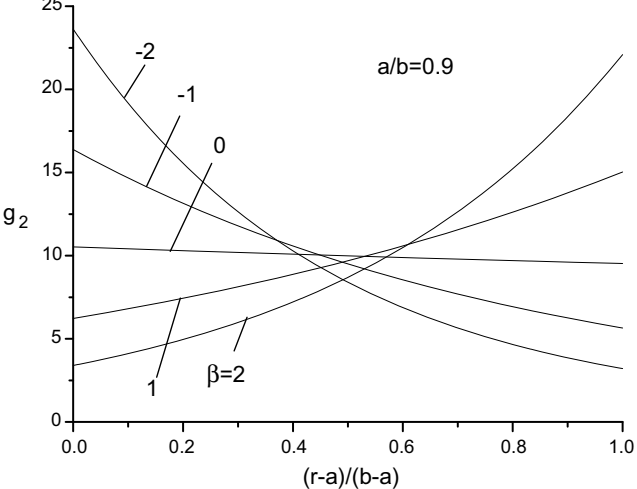
**Fig. 6.** Non-dimensional radial stress  $g_1(a/b, \beta, r)$ , the  $\sigma_r$  component in cylinder ( $a/b = 0.5$ , with a loading  $q_b$  at outer boundary).



**Fig. 8.** Non-dimensional radial stress  $g_1(a/b, \beta, r)$ , the  $\sigma_r$  component in cylinder ( $a/b = 0.9$ , with a loading  $q_b$  at outer boundary).



**Fig. 7.** Non-dimensional tangential stress  $g_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in cylinder ( $a/b = 0.5$ , with a loading  $q_b$  at outer boundary).



**Fig. 9.** Non-dimensional tangential stress  $g_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in cylinder ( $a/b = 0.9$ , with a loading  $q_b$  at outer boundary).

In the spherical pressure vessel case, it is assumed that the Poisson's ratio  $\nu (=0.3)$  takes a constant value, and the Young's modulus has the property [4]

$$E(r) = E_0 \exp\left(\frac{\beta(r-a)}{b-a}\right) \quad \text{with} \quad E(r)|_{r=a} = E_0, \quad E(r)|_{r=b} = E_0 \exp(\beta) \quad (26)$$

where  $E_0$  and  $\beta$  have same meaning as in the previous case.

In this case, the stress-strain relations will be

$$\varepsilon_r = \frac{1}{E(r)}(\sigma_r - 2\nu\sigma_\theta), \quad \varepsilon_\theta = \frac{1}{E(r)}((1-\nu)\sigma_\theta - \nu\sigma_r) \quad (27)$$

Substituting Eq. (25) into (27), and then (27) into (23) yields

$$\frac{d^2 F}{dr^2} - \frac{F}{r^2} - \left(\frac{dF}{dr} - \frac{2\nu}{1-\nu} \frac{F}{r}\right) \frac{1}{E(r)} \frac{dE(r)}{dr} = 0 \quad (\text{or } \Pi(F(r)) = 0) \quad (28)$$

Two boundary value problems are studied below.

In the first problem, the spherical pressure vessel is subject to an inner pressure  $q_a$  and the boundary conditions take the following form

$$\sigma_r|_{r=a} = -q_a, \quad \sigma_r|_{r=b} = 0 \quad (\text{or } F|_{r=a} = -q_a a^2, \quad F|_{r=b} = 0) \quad (29)$$

Similar procedure used in the cylinder case can be used in the present case. The calculated results for stresses can be expressed as

$$\sigma_r = -f_1(a/b, \beta, r) q_a, \quad \sigma_\theta = f_2(a/b, \beta, r) q_a \quad (30)$$

The calculated results for two cases,  $a/b = 0.5$  and  $0.9$  are plotted in Figs. 10–13, respectively.

In the homogeneous material case ( $\beta = 0$ ,  $E(r) = E_0$ ), the available solution is [7–9]

$$\sigma_r = -\frac{(b^3 - r^3)a^3}{(b^3 - a^3)r^3} q_a, \quad \sigma_\theta = \frac{(b^3 + 2r^3)a^3}{2(b^3 - a^3)r^3} q_a \quad (31)$$

From the plotted results in Figs. 10–13, we see that the material parameter  $\beta$  can significantly affects the stress distribution along the radius direction. For example, in the case of  $a/b = 0.5$  and the homogeneous material ( $\beta = 0$  or  $E(r) = E_0$ ), the non-dimensional stresses  $f_2$  (the  $\sigma_\theta$  component) are changed from 0.714, 0.572, ... to 0.214, for  $(r-a)/(b-a) = 0, 0.1, \dots$  to 1.0, respectively (Fig. 11, the curve for  $\beta = 0$ ). However, in the case of  $\beta = 1$ , or  $E(b)/E(a) = 2.7183$ , the non-dimensional stresses  $f_2$  (the  $\sigma_\theta$  component) are varying within the range 0.393–0.310, nearly uniform distribution along the section (Fig. 11, the curve for  $\beta = 1$ ).

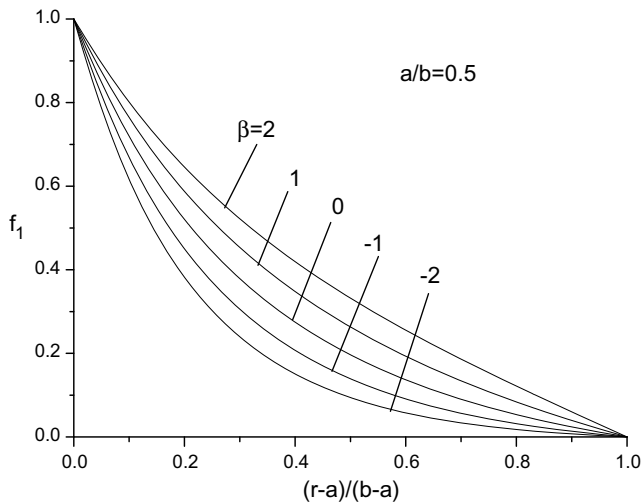


Fig. 10. Non-dimensional radial stress  $f_1(a/b, \beta, r)$ , the  $\sigma_r$  component in spherical pressure vessel ( $a/b = 0.5$ , with an inner pressure  $q_a$ ).

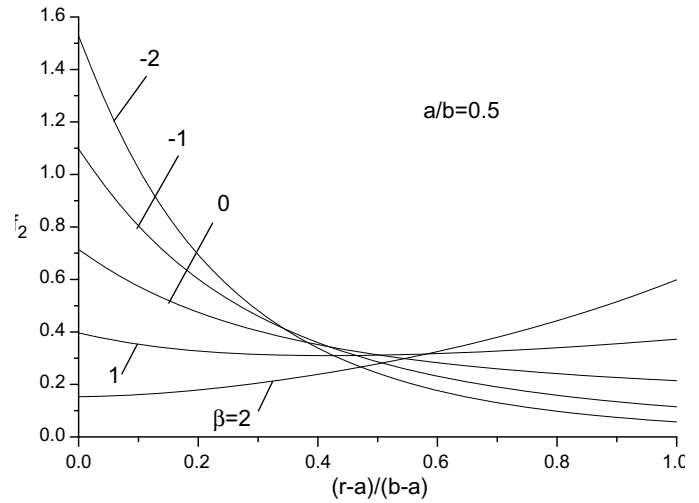


Fig. 11. Non-dimensional tangential stress  $f_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in spherical pressure vessel ( $a/b = 0.5$ , with an inner pressure  $q_a$ ).

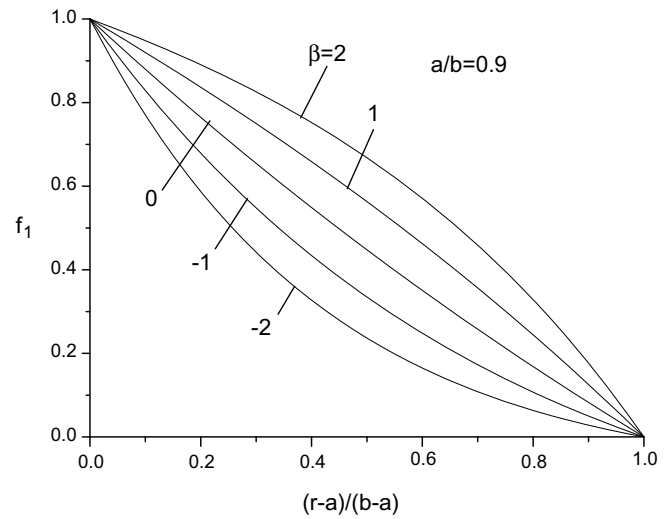


Fig. 12. Non-dimensional radial stress  $f_1(a/b, \beta, r)$ , the  $\sigma_r$  component in spherical pressure vessel ( $a/b = 0.9$ , with an inner pressure  $q_a$ ).

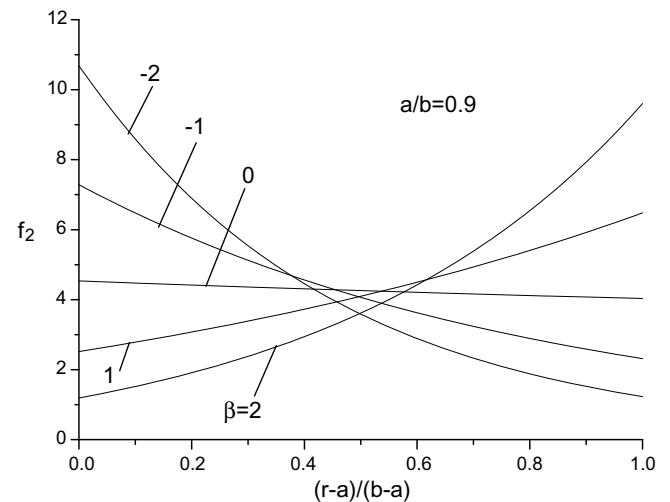


Fig. 13. Non-dimensional tangential stress  $f_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in spherical pressure vessel ( $a/b = 0.9$ , with an inner pressure  $q_a$ ).

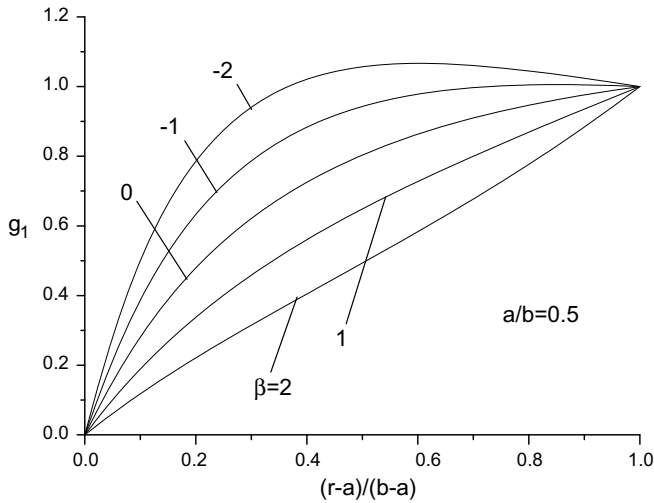


Fig. 14. Non-dimensional radial stress  $g_1(a/b, \beta, r)$ , the  $\sigma_r$  component in spherical pressure vessel ( $a/b = 0.5$ , with a loading  $q_b$  at outer boundary).

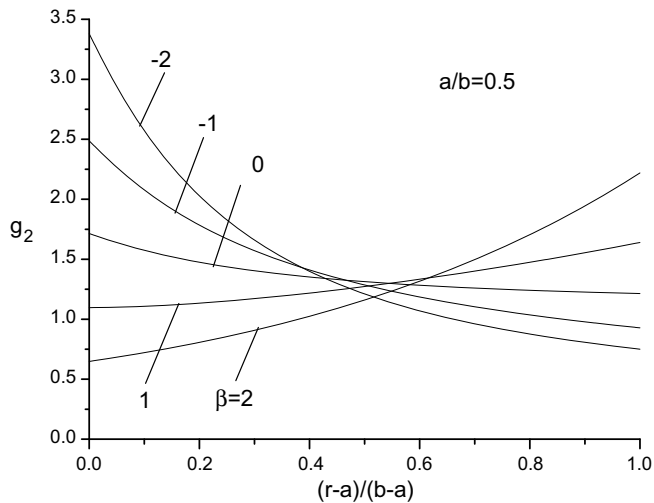


Fig. 15. Non-dimensional tangential stress  $g_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in spherical pressure vessel ( $a/b = 0.5$ , with a loading  $q_b$  at outer boundary).

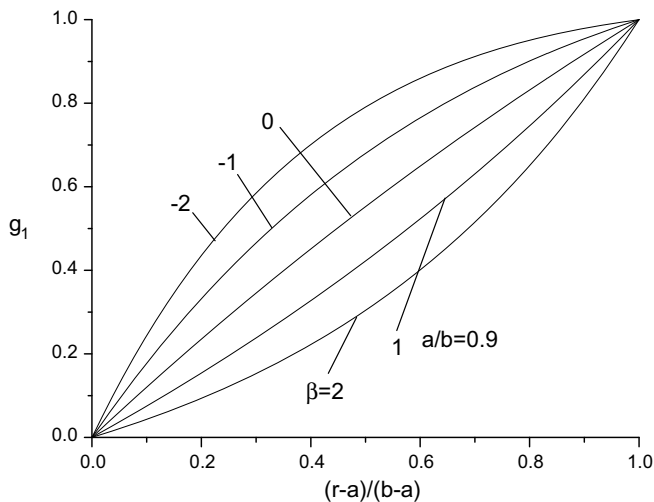


Fig. 16. Non-dimensional radial stress  $g_1(a/b, \beta, r)$ , the  $\sigma_r$  component in spherical pressure vessel ( $a/b = 0.9$ , with a loading  $q_b$  at outer boundary).

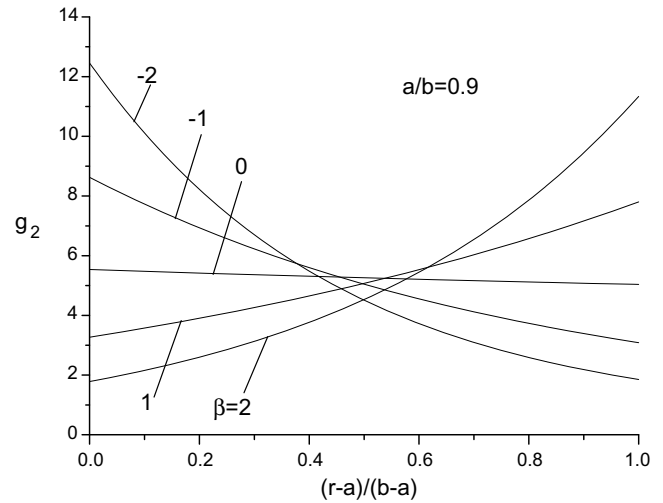


Fig. 17. Non-dimensional tangential stress  $g_2(a/b, \beta, r)$ , the  $\sigma_\theta$  component in spherical pressure vessel ( $a/b = 0.9$ , with a loading  $q_b$  at outer boundary).

In the second problem, the spherical pressure vessel is subject to a loading  $q_b$  at outer boundary, and the boundary conditions take the following form

$$\sigma_r|_{r=a} = 0, \quad \sigma_r|_{r=b} = -q_b \quad (\text{or } F|_{r=a} = 0, \quad F|_{r=b} = -q_b b^2) \quad (32)$$

Similar procedure used in the cylinder case can be used in the present case. The calculated results for stresses can be expressed as

$$\sigma_r = -g_1(a/b, \beta, r)q_b, \quad \sigma_\theta = -g_2(a/b, \beta, r)q_b \quad (33)$$

The calculated results for two cases,  $a/b = 0.5$  and  $0.9$  are plotted in Figs. 14–17, respectively.

In the homogeneous material case ( $\beta = 0$ ,  $E(r) = E_0$ ), the available solution is [7–9]

$$\sigma_r = -\frac{(r^3 - a^3)b^3}{(b^3 - a^3)r^3}q_b, \quad \sigma_\theta = -\frac{(a^3 + 2r^3)b^3}{2(b^3 - a^3)r^3}q_b \quad (34)$$

From the plotted results in Figs. 14–17, we see that the material parameter  $\beta$  can significantly affects the stress distribution along the radius direction. For example, in the case of  $a/b = 0.9$  and the homogeneous material ( $\beta = 0$  or  $E(r) = E_0$ ), the non-dimensional stresses  $g_2$  (the  $\sigma_\theta$  component) are varying within the range 5.035–5.535, nearly uniform distribution along the section (Fig. 17, the curve for  $\beta = 0$ ). However, in the case of  $\beta = 1$ , or  $E(b)/E(a) = 2.7183$ , the non-dimensional stresses  $g_2$  (the  $\sigma_\theta$  component) are changed from 3.270, 3.573, ... to 7.801, for  $(r - a)/(b - a) = 0, 0.1, \dots$  to 1.0, respectively (Fig. 17,  $\beta = 1$ ). In this case, the outer portion is more rigid and the non-dimensional stress at outer portion is higher,  $g_2 = 7.801$  (Fig. 17, the curve for  $\beta = 1$ ).

#### 4. Conclusions

In the homogeneous medium case, for a given loading the stress distribution in an elastic body will be determined uniquely. That is to say there is no approach to adjust the stress distribution in the elastic body even though some portion of the body is subject to a lower level of stress. The use of FGMs can change this inconvenient situation. In this study, it found that the stress distribution in a cylinder or spherical pressure vessel is significantly changed if FGMs is used. For the cylinder or spherical pressure vessel case, an effective numerical method is suggested to evaluate the stress distribution in the medium. The calculated results will help engineer to design a pressure vessel made of FGMs with a reasonable distribution of stress.

## Acknowledgement

The project was supported by National Natural Science Foundation of China.

## References

- [1] J. Tang, H.S. Shen, *Int. J. Nonlinear Mech.* 38 (2003) 467–482.
- [2] P. Gu, R.J. Asaro, *Int. J. Solid Struct.* 34 (1997) 1–17.
- [3] Y.S. Chan, G.H. Paulino, A.C. Fannjiang, *Int. J. Solid Struct.* 38 (2001) 2989–3005.
- [4] F. Erdogan, B.H. Wu, *J. Appl. Mech.* 64 (1997) 449–456.
- [5] M. Jabbari, S. Sohrabpour, M.R. Eslami, *Int. J. Press. Ves. Piping* 79 (2002) 493–497.
- [6] L.H. You, J.J. Zhong, X.Y. You, *Int. J. Press. Ves. Piping* 82 (2005) 347–354.
- [7] S.P. Timoshenko, J.N. Goodier, *Theory of Elasticity*, McGraw-Hill, New York, 1970.
- [8] I.S. Sokolnikoff, *Mathematical Theory of Elasticity*, McGraw-Hill, New York, 1956.
- [9] A.I. Lur'e, *Three-dimensional Problems of the Theory of Elasticity*, Interscience, New York, 1964.
- [10] F.B. Hildebrand, *Introduction to Numerical Analysis*, McGraw-Hill, New York, 1974.